

Pratyush Singh

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Education

Indian Institute of Information Technology, Design & Manufacturing
B.Tech, Computer Science Engineering; CGPA: 7.6/10.0

Jabalpur, India
Aug. 2023 – May 2027

Bhartiyam International School
Higher Secondary Certificate, CBSE; 89.45%

Rudrapur, Uttarakhand
2022

Projects

Expenso – Expense Splitter

[Live](#) | [GitHub](#)

Django REST Framework, React 19, PostgreSQL, Redis, Celery, Docker, Stripe API, Claude API

- Architected a full-stack expense-splitting platform using Django REST Framework, React 19, PostgreSQL, Redis, and Celery, achieving 99% test coverage across 275 backend and 117 frontend tests
- Engineered an $O(n \log n)$ debt-simplification algorithm in Python with Celery-powered background jobs and PostgreSQL query optimization to minimize settlement transactions
- Integrated Stripe payment gateway, Google OAuth, and the Claude API for AI-driven categorization, and deployed 33 REST endpoints via Docker containerization and CI/CD on free-tier hosting at zero infrastructure cost

CollegeBazaar

[Live](#) | [GitHub](#)

React, Express.js, PostgreSQL, JWT, Google OAuth, Cloudinary, Node-cron

- Developed a full-stack campus marketplace using React.js, Express.js, PostgreSQL, JWT authentication, and Google OAuth, powering 32 REST endpoints across 6 modules
- Designed a transaction-safe bidding system using PostgreSQL row-level locking and rollback logic, eliminating race conditions across 50+ concurrent bids
- Implemented automated product-status scheduling with node-cron and secure OTP/Google OAuth authentication, cutting manual campus-marketplace coordination time by roughly 70%

Voice AI Finance Agent

[GitHub](#)

Python, FastAPI, LangGraph, Whisper, FAISS, SQLAlchemy, PostgreSQL, WebSockets

- Built a Voice AI Finance Agent using Python, FastAPI, and Whisper speech-to-text, handling 50+ invoices and expense entries through a voice interface
- Created a multi-agent system using LangGraph and prompt engineering, with 4 specialized agents for finance, research, and task automation
- Combined TTS, FAISS-based vector search for RAG, and PostgreSQL persistence, cutting manual data entry time by nearly 35%

Technical Skills

- **Programming Languages:** C++, Python, Java, JavaScript, TypeScript
- **Frontend:** React.js, Next.js, HTML5, CSS3
- **Backend:** Django, FastAPI, Express.js, REST APIs, JWT, OAuth 2.0, API Integration, Authentication, Authorization
- **Databases:** PostgreSQL, MySQL, MongoDB, Supabase, Redis
- **Cloud & DevOps:** AWS, Docker, Git, GitHub, CI/CD
- **AI/ML:** LLMs, RAG, Vector Databases, Claude API
- **Core Concepts:** OOP, DSA, DBMS, Operating Systems, Computer Networks, System Design

Coding Profiles

- **LeetCode:** pratyush4000 – rating 1704
- **Codeforces:** pratyush40 – Pupil, highest 1289
- **CodeChef:** pratyush400 – 3-star

Achievements

- **Sports Committee Member, IIITDM Jabalpur:** Lawn Tennis Club Representative – organized sports fest "Enthuse" with 500+ participants across 12 categories; coordinated match scheduling, venue logistics, and equipment for weekly club sessions
- **Project Lead – Fusion Project, IIITDM Jabalpur:** Lead developer for institute's official website management system – led ongoing development and maintenance of the platform, serving 2000+ students and faculty; managed updates, bug fixes, and feature releases
- **Competitive Programming:** Solved 800+ problems across multiple competitive programming platforms
- **Contest Performance:** 12th place in Newbie Programming Contest (200+ participants)
- **Certifications:** NPTEL ELITE "Joy of Computing Using Python" – 82% (Top 5% nationwide)